

Osaid Khan

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EDUCATION

Pace University

Master of Science (MS) in Computer Science (GPA - 4.0/4.0)

New York City, NY

Jan 2024 - Dec 2025

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology (B.Tech) in Computer Science

Noida, India

Jul 2016 - Sep 2020

TECHNICAL SKILLS

AI/Agents: LangGraph, LangChain, FastMCP, LangSmith, Langfuse, LlamaIndex, Claude Code, Codex

Backend & Frameworks: FastAPI, Spring Boot, React

Languages: Python, JavaScript/TypeScript, Go, Java

Databases: PostgreSQL, Redis, MongoDB

AWS Services: Step Functions, Lambda, API Gateway, CloudWatch, S3, RDS, EC2, CloudFormation, CloudFront

Others: Docker, Kubernetes, CI/CD, Git

PROFESSIONAL EXPERIENCE

Software Engineer

Phoenix, AZ

Sperse

Feb 2026 - Present

- Designed a production platform for **scalable agentic systems** using **large language models (LLMs)**, **LangGraph**, **LangChain**, **Langfuse**, and **FastAPI**, translating **product requirements** from **cross-functional teams** into multi-step CRM workflows across **40+** specialized sub-agents.
- Built **tool use** and **structured outputs** infrastructure by refactoring **500+** backend operations into typed tool-group dispatchers, replacing **60+** per-agent **REST** tools with validated interfaces that retrieve context, call APIs, and execute actions.
- Developed dynamic planning and tool-routing behavior that exposes only task-relevant tool groups, balancing **latency**, **cost**, **accuracy**, and **controllability** while reducing context bloat and improving tool-selection accuracy by **90%**.
- Implemented **failure handling** and tenant-aware execution through lazy-loaded sub-agent skills, JWT/tenant/role propagation, two-stage input safety with jailbreak regex checks and **Lakera Guard**, and **SQLite** checkpointing for multi-turn state.
- Built **monitoring and observability** plus evaluation infrastructure with trace-driven eval suites and real-time monitors replaying production conversations across prompts and model versions, flagging unsafe agent behavior before release.

AI Researcher

New York City, NY

Pace Artificial Intelligence Lab

Sep 2025 - Dec 2025

- Built an agent-evaluation pipeline for **computer vision and machine learning (CVML)** experiments, ensembling **LLM judges** and reward models to evaluate **multi-step workflows**, rank tool-use runs, and improve agentic system **quality**, **reliability**, and **cost** tradeoffs.
- Implemented a **LangGraph** eval pipeline that scored agent runs, selected high-performing traces, and tuned prompts with **adaptive**, **LLM-driven approaches**, increasing tool-call success by **50%**.

Software Engineer

Noida, India

Qualitest

Feb 2021 - Dec 2023

- Orchestrated long-running production workflows with **AWS Step Functions** and **AWS Lambda**, adding retry/error-recovery paths that improved **failure handling** across **100K+** weekly executions.
- Built end-to-end **CI/CD** pipelines with **GitHub Actions** for production environments, reducing manual deployment steps and improving developer velocity across services, data pipelines, and integrations.
- Optimized **PostgreSQL** queries with window functions, materialized views, and indexes to minimize latency and improve scalable **data systems** behind API-driven workflows.

PROJECTS

Dark Factory - orchestrator for PI agent | *Elixir, TypeScript*

github.com/ozzyozbourne/dark-factory

- Built a distributed coding-agent orchestrator for **LLM-assisted coding** that breaks large requests into dependency-aware task DAGs and executes them across multi-machine AI agents with retries and timeout handling.
- Modeled end-to-end **agentic workflows** with explicit planning, tool execution, and failure handling, contrasting deterministic pipelines with adaptive orchestration for reliable multi-step software tasks.